

Leaf Count Estimation

Single-Image Plant Phenotyping

PeaceOfCode Hackathon — Track 1(a)

PeaceOut Team

Acknowledgement and disclaimer

All mistakes (if any) are ours.

We have used several other sources which we have referred to in the appropriate places.

Problem Statement

Given a single image of a plant I_θ captured at angle θ , learn a function f such that

$$f(I_\theta) = y, \quad y \in \mathbb{N}$$

where y represents the leaf count. Models must be robust to viewpoint changes and partial visibility, and operate under a **single-image inference** setting at test time.

An Important Approach: Multi-View Aggregation (Considered, Not Used)

Idea: Instead of relying on a single viewpoint, aggregate features from all available angles $\{\theta_1, \theta_2, \dots, \theta_k\}$ of the same plant into a single prediction:

$$f(\{I_{\theta_i}\}_{i=1}^k) = y$$

Why it is appealing:

- ▶ Directly resolves occlusion — a leaf hidden at 0° is visible at 90° .
- ▶ Provides richer geometric context; reduces viewpoint-induced variance.

Why we did not use it:

- ▶ The competition evaluation is **strictly single-image inference** — test images are provided one at a time with no angle metadata.
- ▶ No guarantee that complementary angles of a test plant are available.
- ▶ A model trained on multi-view inputs cannot generalise to single-view test conditions without architectural changes.

Key Challenges in Plant Phenotyping

Occlusion Leaf-on-leaf hiding, self-occlusion, and early-stage clustering (days 1–5) cause significant under- or over-counting.

Growth Stage Late-stage density (days 30–40) in Mustard/Wheat; volunteer weeds; ambiguous germination on day 0.

Morphology Compound leaves (Mustard, Radish) — count whole leaf or leaflets? Wheat tillers blur the boundary between leaf and new stalk.

Six Approaches Evaluated

Approach	Backbone	Key Innovation	Val MAE
A1 Baseline	EfficientNet-B3	Crop embed + TTA	0.243
A2 Multi-Task	EfficientNetV2-S	384px + Aux crop head (dropped)	0.217
A3 DINOv2-ViT	DINOv2-Small	Ordinal loss + SWA	0.259
A4 HSV+CNN	EfficientNetV2-S	HSV mask $\rightarrow$ masked RGB	0.258
A5 Per-Crop	EfficientNetV2-S	4 regression heads + 4-channel	0.482
A6 Direct	EfficientNet-B4	L1 loss, clean ablation	0.622

Approaches 1 & 2 — EfficientNet Baseline and High-Res Multi-Task

A1 uses EfficientNet-B3 (224px) with a crop-type embedding concatenated to image features. MAE+MSE compound loss, gradient clipping, and TTA at inference. Val MAE: 0.243.

A2 upgrades to EfficientNetV2-S at 384×384. An auxiliary crop classifier (weight 0.3) is trained jointly but *dropped at inference*, so the backbone learns crop-aware features implicitly — no crop label needed at test time. 8-pass TTA. Val MAE: **0.217** (best overall).

Key lesson: implicit crop guidance via auxiliary training outperforms explicit embedding that leaks label dependency to inference.

Approaches 3 & 4 — ViT/DINOv2 and HSV Segmentation

A3 replaces the CNN backbone with DINOv2-Small (ViT), which excels at patch-level occlusion reasoning. Key additions: ordinal regression loss (soft count labels prevent cliff errors), sinusoidal day embedding, SWA over the last 10 epochs, and 12-pass TTA. Val MAE: 0.259.

A4 uses classical preprocessing: per-crop HSV segmentation thresholds extract a foreground mask which is passed alongside the RGB image into EfficientNetV2-S (last 3 blocks unfrozen). No crop label needed at inference. Val MAE: 0.258.

Both are competitive with A2. A3 shows ViT power; A4 shows domain knowledge value. HSV thresholds remain brittle to lighting/crop colour variation.

Approaches 5 & 6 — Per-Crop Heads and Direct Regression Baseline

A5 uses a union HSV mask (all 4 crop ranges OR'd) for inference independence, feeds 4-channel input (RGB + mask) into EfficientNetV2-S, and routes through 4 separate regression MLPs (one per crop) with soft-weighted averaging. Cosine LR with warmup. Val MAE: 0.482 — significant overfitting observed despite the richest pipeline.

A6 is a clean ablation: EfficientNet-B4 at 384px, single regression head, L1 loss, AdamW, ReduceLROnPlateau. No auxiliary tasks or segmentation. Val MAE: 0.622 — confirms that auxiliary crop guidance is critical.

Complexity does not guarantee better results; A5's overfit suggests insufficient data for full per-crop specialisation.

Conclusion

- ▶ **Best approach:** A2 (Val MAE 0.217) — EfficientNetV2-S at 384px with implicit multi-task crop guidance.
- ▶ **Resolution matters:** 224px $\rightarrow$ 384px consistently helps for small-leaf detail and early-growth stages.
- ▶ **Classical preprocessing is competitive:** HSV masking (A4) reaches 0.258 without any test-time label dependency.
- ▶ **Complexity $\neq$ performance:** A5 (most complex) overfit; A3 (ViT + ordinal) limited by dataset size.
- ▶ **Multiview:** The strongest theoretical approach — but incompatible with the single-image inference requirement.
- ▶ **Path forward:** Ensemble A2 + A4; cross-crop augmentation; larger dataset to unlock A3-class architectures.